

Product Requirements Document (PRD)

Product: Paws Off Bear Deterrent Spray

Version: 1.0

Date: May 6, 2026

1. Product Vision

Paws Off is a premium bear deterrent spray designed for the U.S. market. The product must balance high-performance safety standards with a modern, approachable brand identity. To expedite time-to-market and reduce regulatory costs, the product will be formulated to qualify for the EPA's B660 "Me-Too" registration pathway, matching the efficacy and safety profile of established market leaders.

2. Regulatory Strategy

The core strategy relies on achieving a "substantially similar" formulation to an existing EPA-registered product (Target: EPA Reg. No. 55541-2) ¹. This approach allows Paws Off to cite existing acute toxicity and efficacy data, bypassing the need for extensive new laboratory testing.

2.1 EPA Clearance Requirements

To secure the B660 registration, the product must meet the following criteria:

- Active Ingredient Match:** Must contain exactly 2.0% capsaicin and related capsaicinoids derived from Oleoresin of Capsicum ¹.
- Inert Ingredient Similarity:** The remaining 98.0% of the formulation must closely mirror the target product's Confidential Statement of Formula (CSF).
- Approved Inerts:** All solvents, propellants, and emulsifiers must be listed on the EPA List 4B (Inert Ingredients of Minimal Concern) ².

3. Formulation Specifications

Based on reverse-engineering of the target product, the formulation must adhere to the following composition ¹:

Component Function	Proposed Ingredient	Required % by Weight
Active Ingredient	Oleoresin Capsicum (yielding exactly 2.0% capsaicinoids)	~2.0%

Propellant	Propane / Isobutane Blend (A-70)	65.0% - 75.0%
Primary Solvent	Isopropyl Alcohol	15.0% - 25.0%
Co-Solvent	Propylene Glycol	2.0% - 5.0%
Emulsifier	Polysorbate 80	1.0% - 3.0%

Note: This formulation will result in an "Extremely Flammable" classification, which is consistent with the target product's current labeling.

4. Hardware & Packaging Requirements

The physical canister and delivery system are critical for both performance and regulatory compliance.

4.1 Canister Specifications

- **Material:** Steel construction for durability in outdoor environments.
- **Style:** Slant-shoulder design.
- **Dimensions:** 180mm (height) x 53mm (diameter).
- **Regulatory Standard:** Must meet U.S. Department of Transportation (DOT) 2Q burst pressure standards to safely contain the hydrocarbon propellant blend.

4.2 Delivery System Performance

The actuator and nozzle assembly must be tuned to the specific viscosity and pressure of the formulation to achieve the following:

- **Spray Pattern:** A heavy, expanding fog designed to create a wide barrier between the user and the animal.
- **Effective Range:** The spray must project a minimum distance of 35 feet (10.6 meters).
- **Discharge Time:** The canister must empty in at least 7 seconds of continuous spraying.
- **Environmental Tolerance:** The system must function reliably in freezing conditions, necessitating a propellant with appropriate vapor pressure at low temperatures.

5. References

[1] Bear Spray Formulation Analysis: Reverse-Engineering Counter Assault.

[2] U.S. Environmental Protection Agency. Inert Ingredients Ordered by CAS Number - List 4B.